

1D CNN — Architecture Variant Comparison

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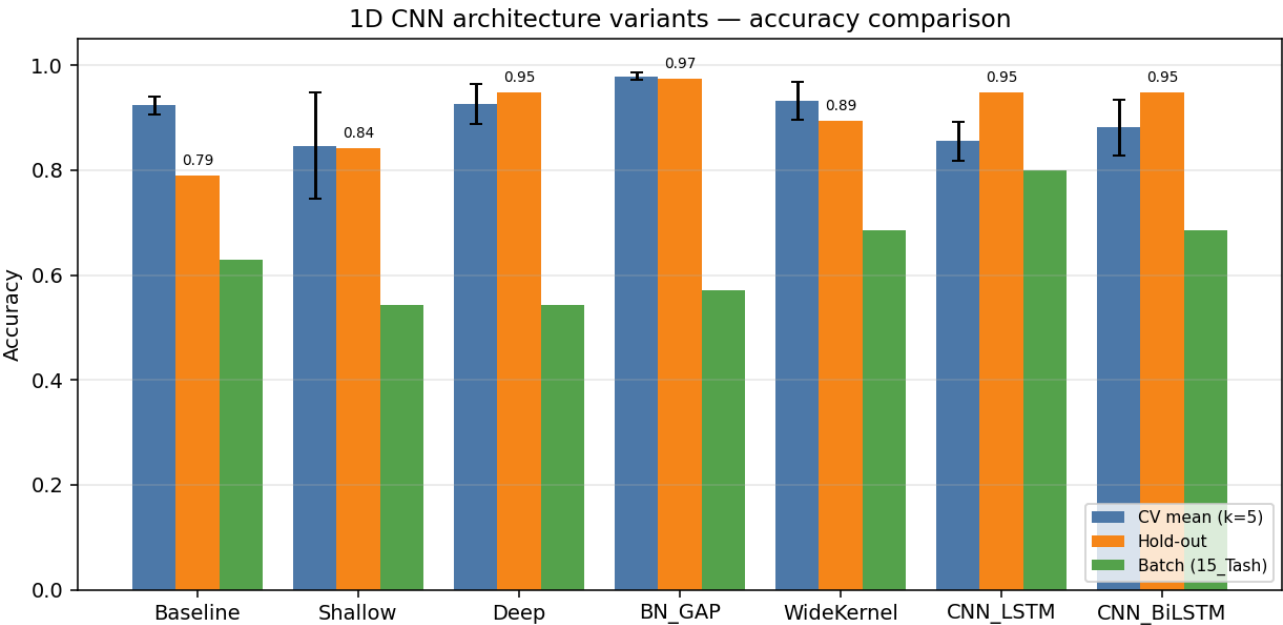
Data & configuration

Training path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic
Training path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic
Training path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic
Training path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic
Training path 5	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic
Training path 6	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic
Training path 7	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic
Training path 8	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic
Training path 9	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic
Training path 10	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic
Training path 11	/home/jestin/ThesisRepo/ML/NewTestData/18_Bridgette/Dynamic
Batch-test root	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Dynamic
Total trials	376 (train pool 338, hold-out 38)
Classes	7: Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	176 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 15.0 Hz, order 4
Normalisation	minmax
CV folds · shuffle	5 · True
Augmentation	on · copies=5 · amplitude_scale, baseline_offset
Epochs · batch size	15 · 16

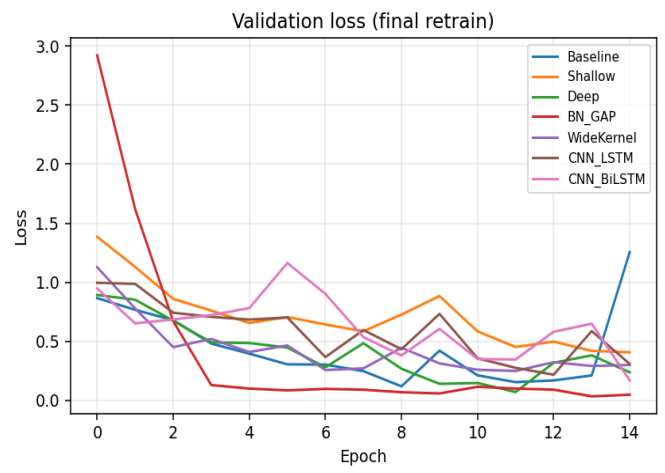
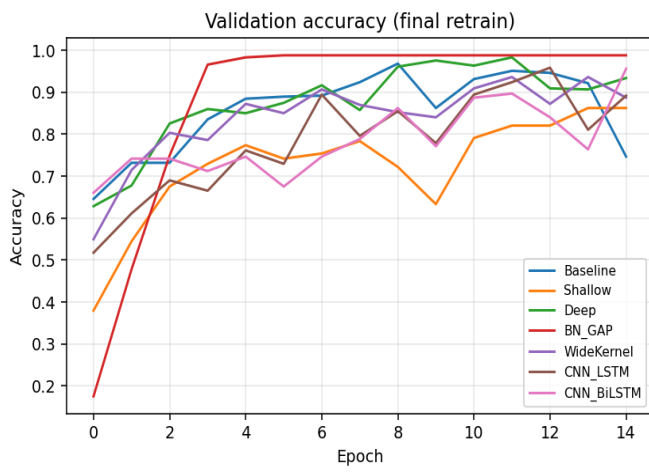
Variant comparison (summary)

Variant	Params	CV mean (k=5)	CV std	Hold-out acc	Hold-out loss	Batch acc	Batch n
Baseline	113,255	0.9232	0.0169	0.7895	1.1509	0.6286	22/35
Shallow	72,487	0.8459	0.1013	0.8421	0.5573	0.5429	19/35
Deep	204,007	0.9259	0.0377	0.9474	0.1618	0.5429	19/35
BN_GAP	57,319	0.9793	0.0073	0.9737	0.2458	0.5714	20/35
WideKernel	131,687	0.9319	0.0359	0.8947	0.2467	0.6857	24/35
CNN_LSTM	68,455	0.8550	0.0369	0.9474	0.2456	0.8000	28/35
CNN_BiLSTM	105,575	0.8816	0.0535	0.9474	0.2894	0.6857	24/35

Best on hold-out: **BN_GAP** · Best on CV: **BN_GAP**



Training curves (final retrain on full training pool)



Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	113,255
CV mean ± std	0.9232 ± 0.0169
Hold-out test acc / loss	0.7895 · 1.1509
Batch-test acc	0.6286 (22/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1620	68	0.9118	0.9853	0.9118
2	1620	68	0.9118	0.9706	0.9118
3	1620	68	0.9118	0.9265	0.9118
4	1626	67	0.9552	0.9552	0.9552
5	1626	67	0.9254	0.9701	0.9254

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	0.714	1.000	0.833	5
Double_Nothing	0.667	0.400	0.500	5
Double_Pistol_Recoil	0.857	1.000	0.923	6
Double_Raise	0.857	1.000	0.923	6
Double_Wiggle	0.625	1.000	0.769	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	0.000	0.000	0.000	5
macro avg	0.674	0.771	0.707	38
weighted avg	0.693	0.789	0.726	38

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	0	5	0.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	2	5	0.400
Drum_Roll	0	5	0.000
Overall	22	35	0.629

Variant: Shallow

Single Conv block: Conv1D(32,k=5)→Pool→Flatten→Dense(32)→Dropout(0.3)→Softmax

Trainable params	72,487
CV mean ± std	0.8459 ± 0.1013
Hold-out test acc / loss	0.8421 · 0.5573
Batch-test acc	0.5429 (19/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1620	68	0.9118	0.9853	0.9118
2	1620	68	0.9706	0.9706	0.9706
3	1620	68	0.7353	0.8235	0.7353
4	1626	67	0.8955	0.9104	0.8955
5	1626	67	0.7164	0.7910	0.7164

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	0.800	0.889	5
Double_Nothing	0.800	0.800	0.800	5
Double_Pistol_Recoil	1.000	0.833	0.909	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	0.556	1.000	0.714	5
Double_cmere	0.857	1.000	0.923	6
Drum_Roll	1.000	0.400	0.571	5
macro avg	0.888	0.833	0.830	38
weighted avg	0.893	0.842	0.839	38

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	2	5	0.400
Double_Nothing	1	5	0.200
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	1	5	0.200
Drum_Roll	0	5	0.000
Overall	19	35	0.543

Variant: Deep

Three Conv blocks: Conv1D(32,4)→Conv1D(64,4)→Conv1D(128,3)→Flatten→Dense(128)→Dropout(0.4)→Softmax

Trainable params	204,007
CV mean ± std	0.9259 ± 0.0377
Hold-out test acc / loss	0.9474 · 0.1618
Batch-test acc	0.5429 (19/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1620	68	0.9559	0.9559	0.9559
2	1620	68	0.9559	0.9853	0.9559
3	1620	68	0.8971	0.9559	0.8971
4	1626	67	0.8657	0.9403	0.8657
5	1626	67	0.9552	0.9701	0.9552

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.600	0.750	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	0.833	1.000	0.909	5
Double_cmere	0.857	1.000	0.923	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.956	0.943	0.940	38
weighted avg	0.956	0.947	0.943	38

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	2	5	0.400
Double_Pistol_Recoil	2	5	0.400
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	0	5	0.000
Drum_Roll	0	5	0.000
Overall	19	35	0.543

Variant: BN_GAP

Conv+BN+ReLU ×3 with GlobalAveragePooling1D head — fewer params, less overfit

Trainable params	57,319
CV mean ± std	0.9793 ± 0.0073
Hold-out test acc / loss	0.9737 · 0.2458
Batch-test acc	0.5714 (20/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1620	68	0.9853	0.9853	0.9853
2	1620	68	0.9853	0.9853	0.9853
3	1620	68	0.9706	0.9853	0.9706
4	1626	67	0.9701	0.9701	0.9701
5	1626	67	0.9851	0.9851	0.9851

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	0.833	1.000	0.909	5
Double_Nothing	1.000	0.800	0.889	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	1.000	1.000	1.000	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.976	0.971	0.971	38
weighted avg	0.978	0.974	0.973	38

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	2	5	0.400
Double_Nothing	3	5	0.600
Double_Pistol_Recoil	3	5	0.600
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	2	5	0.400
Drum_Roll	0	5	0.000
Overall	20	35	0.571

Variant: WideKernel

Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	131,687
CV mean ± std	0.9319 ± 0.0359
Hold-out test acc / loss	0.8947 · 0.2467
Batch-test acc	0.6857 (24/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1620	68	0.9118	0.9412	0.9118
2	1620	68	0.9853	0.9853	0.9853
3	1620	68	0.9265	0.9265	0.9265
4	1626	67	0.8806	0.9104	0.8806
5	1626	67	0.9552	0.9701	0.9552

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.600	0.750	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	0.667	1.000	0.800	6
Double_Wiggle	0.833	1.000	0.909	5
Double_cmere	1.000	0.667	0.800	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.929	0.895	0.894	38
weighted avg	0.925	0.895	0.892	38

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	3	5	0.600
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	0	5	0.000
Drum_Roll	1	5	0.200
Overall	24	35	0.686

Variant: CNN_LSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→LSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	68,455
CV mean ± std	0.8550 ± 0.0369
Hold-out test acc / loss	0.9474 · 0.2456
Batch-test acc	0.8000 (28/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1620	68	0.8088	0.9559	0.8088
2	1620	68	0.8676	0.9118	0.8676
3	1620	68	0.9118	0.9118	0.9118
4	1626	67	0.8209	0.8358	0.8209
5	1626	67	0.8657	0.8657	0.8657

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.600	0.750	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	0.833	1.000	0.909	5
Double_cmere	0.857	1.000	0.923	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.956	0.943	0.940	38
weighted avg	0.956	0.947	0.943	38

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	3	5	0.600
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	4	5	0.800
Drum_Roll	1	5	0.200
Overall	28	35	0.800

Variant: CNN_BiLSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→BiLSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	105,575
CV mean ± std	0.8816 ± 0.0535
Hold-out test acc / loss	0.9474 · 0.2894
Batch-test acc	0.6857 (24/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1620	68	0.9118	0.9412	0.9118
2	1620	68	0.8529	0.9559	0.8529
3	1620	68	0.9118	0.9265	0.9118
4	1626	67	0.9403	0.9403	0.9403
5	1626	67	0.7910	0.9552	0.7910

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.600	0.750	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	0.857	1.000	0.923	6
Double_Wiggle	0.833	1.000	0.909	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.956	0.943	0.940	38
weighted avg	0.956	0.947	0.943	38

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	2	5	0.400
Double_Pistol_Recoil	3	5	0.600
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	2	5	0.400
Drum_Roll	2	5	0.400
Overall	24	35	0.686